

Panda: Continued Investigation

Mechanism Search, Retractions, and the Path Ahead

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Outline

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Since We Last Met: What Held Up

Confirmed, not just repeated:

- Weather advantage: replicated independently multiple times, now at $p < 0.001$ via 3 separate routes
- Channel attention: not the driver (was 2 nulls, now 5)
- Non-chaotic Burgers advantage: representation-robust (3 encodings)

Proposed directions – executed:

- “Koopman lifting ablation” → became **A1**: full 100k campaign
- “Persistent homology” → became the **TDA investigation**

New since last time

[OBS] Seasonal-naïve beats *both* Panda and Chronos on ETTh2 (H=96). A trivial baseline, no learned parameters. **[HYP]** Reframes “Panda fails to generalize to ETTh” as “ETTh favors simple models entirely.”

Since We Last Met: What Didn't Hold Up

Claim (last meeting)	What happened	Status
Sensor heterogeneity: <i>"most controlled finding in the investigation"</i>	Collapsed 83% $\rightarrow$ 4% at $n=20$. Confirmed twice, independently.	[RETIRED] Retired
Harmonic oscillator: key evidence <i>against</i> chaos-necessity	Amplitude-growth generator artifact. Advantage collapses 98.8% on stable generator.	[RETRACTED] Re- tracted
ETTh2 H=336: +0.185, $p = 0.013$	Independent rerun: +0.061, $p = 0.088$. Was flagged exploratory at the time – this is the caution being vindicated.	[RETIRED] Non- replicated

Why this matters: two of these directly motivated proposed next steps from the last meeting (G-SWaN node embeddings depended on heterogeneity). That motivation is now gone.

Confidence Tiers Today: High & Medium

High confidence:

- Weather advantage ($n=20$, $p \leq 0.001$)
- Non-chaotic Burgers advantage
- Channel attention not the driver (5 nulls)
- **Lorenz phase-surrogate** (next slide)
- Prediction head not a cheap fix
- Full-scale univariate confirmation

Medium confidence:

- A1's Koopman-ablation pattern (real, mechanism unexplained)
- A3: lift geometry doesn't support the theory
- A2a: temporal attention, weak evidence
- G1: rollout-structure preservation (thin base, 2 systems)
- B3a: periods-in-window, replicated lead
- B3b: Burgers representation, robust at $\nu=0.05$

Confidence Tiers Today: Retired & Retracted

[RETIRED] Retired (tested, closed):

- Sensor heterogeneity bottleneck
- Intrinsic attractor geometry (Branch B1)
- Permutation entropy / naive correlation dimension
- Node-identity scalar-offset proxy
- Resolution-dependency (B3c)
- Distributional similarity to skew40
- ETTh2 H=336

[RETRACTED] Retracted (was wrong):

- Harmonic oscillator advantage

As it stands

Five consecutive candidate mechanisms tested (channel attention, Koopman geometry, temporal attention, resolution, distributional similarity). **None** cleanly explains the advantage. The advantage itself is untouched.

The Flagship Result: Lorenz Phase-Surrogate

Question: does Panda's advantage need genuine deterministic spectrum, wildly different sensitivity. structure, or just signal statistics (variance, spectrum)?

Method: phase-randomize Lorenz ($\rho=28$) – preserves the power spectrum *exactly*, destroys all temporal/phase structure

Condition	Panda	Chronos	Adv. (p)
Chaotic (real)	0.050	0.534	+0.484 ($<0.0001^*$)
Surrogate	0.784	1.036	+0.252 ($<0.0001^*$)

$n = 20$. Confirmed, corrected significance framing.

[OBS] Panda degrades $\approx 15.7\times$.

Chronos degrades $\approx 1.9\times$. Same

structure, or just signal statistics (variance, spectrum)?

[PAT] (High confidence.)

highest-confidence single result in the

log: Panda specifically exploits

deterministic structure, not

second-order statistics.

This motivates the rest of this

section: we know Panda *needs* this

structure. What we don't know is

which component provides the

sensitivity.

A1: Koopman Lifting Ablation (100k Campaign)

Method: retrain from scratch with the fixed random-feature lift replaced by a plain linear projection. Full campaign: 3 Lorenz protocols, 2 held-out systems (Rossler, SprottB), Bonferroni-corrected significance, OOD sweep.

System class	Winner	Sig.
Chaotic ODEs	Ablation	5/6 Bonferroni
Burgers $\nu=1.0$	Baseline	Yes

[PAT] (High confidence.) A real, replicated behavioral pattern. Fixed lift helps on chaotic ODEs, hurts on Burgers.

But: the theoretical account offered for *why* (Koopman/eDMD linearization) was never tested directly until A3, next slide.

Confound acknowledged: $4\times$ smaller batch than published (256 vs. 1024), hardware-forced. Affects absolute magnitude, not the baseline-vs-ablation direction.

A3: Does the Lift Actually Do What the Theory Says?

Question: A1 found a pattern. Is it explained by Koopman/eDMD linearization, as the architecture's own stated motivation suggests?

Two independent tests on the fixed lift's own geometry:

- ① eDMD reconstruction residual (does a linear operator fit the lift's features well?)
- ② Jacobian sensitivity (does the lift track chaotic sensitivity differently than non-chaotic?)

[OBS] Burgers is the **worst**-fit class for a linear operator, not the best – exactly backwards from what the theory predicts.

[OBS] Jacobian sensitivity shows **no separation** between chaotic and non-chaotic classes.

[PAT] (High confidence.) Neither test supports the theoretical account. A1's pattern is real; its explanation is not this.

A2a: Temporal Attention Probes

Question: is temporal (cross-patch) attention the missing piece? Tested cheaply, at inference time, no retrain.

Three probes:

- ① Patch-order shuffling (the pre-registered gate)
- ② Context-length truncation
- ③ Attention-map inspection (descriptive)

[OBS] Probe 1's literal gate says ESCALATE; a natural scale-correction reverses it to DO NOT ESCALATE. Not robust to that choice.

[OBS] Probe 2's asymmetry (Burgers improves with less context) is reproduced almost exactly by **non-learned linear baselines** – no attention mechanism required.

[PAT] (Medium confidence.) Weak, partly self-undermining. A2b (full retrain) not pursued – deprioritized, not rejected.

Structure Statistic Investigation (Persistent Homology)

The executed version of last meeting's "persistent homology" proposal.

Built properly: 3 rounds of gate validation before touching real data, after two prior estimator failures (Rosenstein λ_1 , naive correlation dimension).

Real-data result: Weather splits cleanly – thermo/humidity channels score high-structure, wind/radiation score low.

[OBS] Downsampling control (matching sampling rate to ETTh): the Weather-vs-ETTh separation **does not survive**. ARTIFACT verdict.

[PAT] (Medium confidence.) The within-Weather split is real and narrowly scoped (“detects smooth rotational structure,” not general topology). The cross-dataset story is retired.

Branch B1 (intrinsic geometry) retired, merged into the broader signal-family question.

Branch B: Nulls, Plus One Active Lead

Candidate	Result
B3c: Resolution	No support. Neither model's MAE changes significantly at hourly Weather.
B3b: Representation	Robust at $\nu=0.05$; mixed but direction-consistent at $\nu=1.0$.
G5: Horizon mismatch	Not supported. Chronos's spec ceiling doesn't explain the Weather result.
Skew40 similarity	Weak, never reaches significance at honest effective n . Closed.

B3a – different in kind

Not a null. A **replicated** (2 seeds), non-monotonic pattern in periods-in-window, with a concrete falsifiable mechanism: $\text{piw}=32$ aligns exactly with Panda's patch size (16).

Active lead, not yet confirmatory.

Headline: Five Nulls, Mechanism Still Open

Channel attention • Koopman geometry •
Temporal attention • Resolution • Distributional similarity

None of these cleanly explains the advantage.

The advantage itself (Weather, non-chaotic Burgers, the Lorenz surrogate sensitivity) is real, replicated, and untouched by any of these nulls.

What this means for the project: we've exhausted the obvious single-component leave-one-out tests. Two honest paths remain: keep searching (joint/multi-component ablations), or **pivot to improving the model directly**. We chose the second.

Lesson 1: The Heterogeneity Collapse (Part 1)

The story as told last meeting: three converging, difficulty-controlled experiments (Exp 24–26) showed Panda MAE nearly doubling from homogeneous to heterogeneous Weather subsets, at $n=8$.

Subset	$n=8$	$n=20$
homo_matched	0.331	0.554
hetero_controlled	0.605	0.561
% change	83%	4%

[OBS] At $H=336$, the effect doesn't just shrink – it **reverses sign** (homo becomes *worse* than hetero).

Confirmed independently, twice: Exp 33 ($n=20$ discriminator) and Exp 34 (separate code path, channel indices verified against source).

Lesson 1: The Heterogeneity Collapse (Part 2)

What was ruled out, one by one:

- Per-call model stochasticity (R1: bit-for-bit deterministic)
- Seasonal window clustering (checked directly, not supported)
- Channel-index mismatch (verified byte-identical to source)

Conclusion: small-sample variance in window selection, not a real architectural property.

Consequence

G-SWaN node embeddings and XXLTraffic evaluation (Branch B2) – **last meeting's Proposed Direction 2** – are retired outright. The effect they were meant to explain doesn't exist.

The lesson: $n=8$ can look like a strong, controlled, converging finding and still be noise. This is now the project's standing caution for every remaining $n=8$ result.

Lesson 2: The Harmonic Retraction (Part 1)

What we told you: Panda beats Chronos on a pure sinusoid (+0.370, later “confirmed” at $n=20$, $p < 0.0001$) – the cleanest evidence against chaos-necessity.

What we found: the Harmonic generator used explicit-Euler integration. For an *undamped* oscillator, this is provably unstable – discrete-map eigenvalue magnitude > 1 for any nonzero ωdt .

[OBS] Within a single 512-sample evaluation window: amplitude grows **1.37×**. Across the full trajectory: **42×**.

This is not a subtle numerical quibble. The signal being evaluated was not the stationary sinusoid it was supposed to be.

Lesson 2: The Harmonic Retraction (Part 2)

Generator	H=96 Adv.	Sig.
Euler (original)	+0.322	$p < 0.0001$
Stable (RK45)	+0.004	marginal
Reduction	98.8%	–

[OBS] Decomposed: Chronos improved $4.1\times$ more than Panda when the generator was fixed. The original advantage was inflated by Chronos's degradation on a runaway signal, not genuine Panda capability.

Consequence, and the checkup

Chaos-not-necessary now rests on non-chaotic Burgers alone. A1/A2a/A3's Harmonic-touching results used the same flawed generator – **re-checked**: A1's pattern survives on the stable generator, smaller effect size, same direction.

Same class of bug, checked elsewhere:

Duffing audited too – see later slide.

G1: Rollout-Structure Preservation

Question: do Panda's *autoregressive rollouts* preserve correct attractor structure, not just pointwise accuracy?

Method: Grassberger-Procaccia correlation dimension – first successful gating of this estimator family in the project's history, validated at both long-N and application-N.

System	Panda err.	Chronos err.	Ratio
Double Pendulum	0.555	1.229	2.2×
Lorenz $\rho=10$	0.080	0.677	8.5×

[PAT] (High confidence.) on these two systems: Panda's rollouts preserve attractor dimension far better than Chronos's.

[HYP] (Medium confidence.) overall – thin evidentiary base (2 systems, 1 trajectory each). The most promising new lead, not yet Tier 1.

Along the way: found and fixed a real Double Pendulum equation bug, and discovered session-level (not call-level) non-determinism in baseline_100k.

Full-Scale Univariate Confirmation

The gap: channel attention was tested 4 times, but never at full 21-channel scale against the real Chronos comparison, matching Experiment 8's exact protocol.

H	Uni. Adv.	p
96	+0.201	$< 0.0001^*$
192	+0.222	0.0003*
336	+0.177	0.0001*

[OBS] (High confidence.) the cleanest significance anywhere in this log. 5th independent channel-attention null, and the strongest.

Consistency-gated against Experiment 8 before interpretation (diff < 0.03 on both models).

Duffing Integrator Audit

Given the Harmonic bug: does the same risk exist elsewhere? Duffing uses the identical manual-Euler coding pattern.

Check	Threshold	Result
Window growth ratio	$\geq 1.3\times$	$1.05\times$
Long-run std deviation	$> 15\%$	3.9%

[OBS] (High confidence.) Both checks pass cleanly. No caveat needed on Duffing's result.

[HYP] (Medium confidence.) Duffing's damping term dissipates the spurious energy Euler injects; Harmonic's undamped case had nothing to counteract it. Predicts which other systems are at risk.

From Explain to Improve: Trainable Koopman Lift

Motivation: A1 found the fixed lift helps chaotic/Harmonic, hurts Burgers. A3 showed the fixed lift's own geometry doesn't do clean Koopman linearization. **Is fixedness itself the problem?**

Design:

- Unfreeze RFF only (keep polynomial features fixed) – surgical, isolates one variable
- 3-arm: baseline / ablation / trainable
- In-distribution gate first, freshly computed and saved

What can happen

- ① Trainable closes the Burgers gap without losing chaotic/Harmonic → fixedness was the problem
- ② Same helps/hurts split persists → not about fixedness, something about RFF+poly structure itself
- ③ Undertrained/inconclusive everywhere

Next Steps?

In progress / ready:

- Trainable Koopman lift
- ERA5 / BARRA-R – fills the real-world-*chaotic* gap (every real dataset tested so far is non-chaotic; every chaotic system tested is synthetic)

Cheap, queued:

- B3a denser grid + second seed (patch-alignment mechanism)
- Joint/multi-component ablation (tests whether the mechanism is diffuse, not localized)

Discussion points

- Feedback on the trainable-lift design
- Priorities: lift vs. ERA5 vs. joint ablation
- Anything the tree surfaces that has been mischaracterized

Thank You

Questions and suggestions welcome.